ARID DESERT LANDFORMS / THAR DESERTIFICATION

CONCEPT FIREWALL → DESERT-FORMING CONTROLS → AEOLIAN TRANSPORT → DUNE DISCRIMINATOR → WATER IN THE DESERT → THAR SPATIAL ANATOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW · ◇ SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 CONCEPT FIREWALL

CONCEPT FIREWALL · CLIMATIC REGION · LONG-TERM MOISTURE DEFICIT · TEMPORARY ANOMALY · DEGRADATION WITHIN DRYLANDS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
DESERT → climatic region	DROUGHT → temporary anomaly	DESERT → climatic region
ARIDITY → long-term moisture deficit	DESERTIFICATION → degradation within drylands	ARIDITY → long-term moisture deficit

DEFINITION / COMPONENTS
- DESERT → climatic region
- ARIDITY → long-term moisture deficit

TRIGGER / MECHANISM
- DROUGHT → temporary anomaly
- DESERTIFICATION → degradation within drylands

EFFECT / INTERVENTION
- DESERT → climatic region
- ARIDITY → long-term moisture deficit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DROUGHT → temporary anomaly; DESERTIFICATION → degradation within drylands.

01

STAGE 01 DESERT-FORMING CONTROLS

DESERT-FORMING CONTROLS · SUBTROPICAL HIGH · DESCENDING DRY AIR · RAIN SHADOW · MOISTURE LOSS · COLD CURRENT · STABLE DRY COASTAL AIR · POLAR COLD

SUBTROPICAL HIGH → descending dry air → CONTINENTALITY / RAIN SHADOW → moisture loss → COLD CURRENT → stable dry coastal air → POLAR COLD → low absolute moisture

DEFINITION / COMPONENTS
- SUBTROPICAL HIGH → descending dry air
- CONTINENTALITY / RAIN SHADOW → moisture loss

TRIGGER / MECHANISM
- COLD CURRENT → stable dry coastal air
- POLAR COLD → low absolute moisture

EFFECT / INTERVENTION
- SUBTROPICAL HIGH → descending dry air
- CONTINENTALITY / RAIN SHADOW → moisture loss

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COLD CURRENT → stable dry coastal air; POLAR COLD → low absolute moisture.

02

STAGE 02 AEOLIAN TRANSPORT

AEOLIAN TRANSPORT · COARSE GRAINS ROLL OR SLIDE · SAND GRAINS BOUNCE · FINE DUST TRAVELS FAR · VELOCITY OR CARRYING POWER FALLS

CREEP → coarse grains roll or slide → SALTATION → sand grains bounce → SUSPENSION → fine dust travels far → DEPOSITION → velocity or carrying power falls

DEFINITION / COMPONENTS
- CREEP → coarse grains roll or slide
- SALTATION → sand grains bounce

TRIGGER / MECHANISM
- SUSPENSION → fine dust travels far
- DEPOSITION → velocity or carrying power falls

EFFECT / INTERVENTION
- CREEP → coarse grains roll or slide
- SALTATION → sand grains bounce

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUSPENSION → fine dust travels far; DEPOSITION → velocity or carrying power falls.

03

STAGE 03 DUNE DISCRIMINATOR

DUNE DISCRIMINATOR · LIMITED SAND, HORNS DOWNWIND · ABUNDANT SAND, ONE DOMINANT WIND · ALIGNED WITH RESULTANT WINDS · VEGETATION ANCHOR · MULTIDIRECTIONAL WINDS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
BARCHAN → limited sand, horns downwind	LONGITUDINAL → aligned with resultant winds	BARCHAN → limited sand, horns downwind
TRANSVERSE → abundant sand, one dominant wind	PARABOLIC / STAR → vegetation anchor / multidirectional winds	TRANSVERSE → abundant sand, one dominant wind

DEFINITION / COMPONENTS
- BARCHAN → limited sand, horns downwind
- TRANSVERSE → abundant sand, one dominant wind

TRIGGER / MECHANISM
- LONGITUDINAL → aligned with resultant winds
- PARABOLIC / STAR → vegetation anchor / multidirectional winds

EFFECT / INTERVENTION
- BARCHAN → limited sand, horns downwind
- TRANSVERSE → abundant sand, one dominant wind

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARABOLIC / STAR → vegetation anchor / multidirectional winds.

04

STAGE 04 WATER IN THE DESERT

WATER IN THE DESERT · STORM RUNOFF · WADI DISCHARGE · MOUNTAIN FRONT · ALLUVIAL FAN · FANS COALESCE · CLOSED BASIN · PLAYA AND SALT CONCENTRATION

STORM RUNOFF → wadi discharge → MOUNTAIN FRONT → alluvial fan → FANS COALESCE → bajada → CLOSED BASIN → playa and salt concentration

DEFINITION / COMPONENTS
- STORM RUNOFF → wadi discharge
- MOUNTAIN FRONT → alluvial fan

TRIGGER / MECHANISM
- FANS COALESCE → bajada
- CLOSED BASIN → playa and salt concentration

EFFECT / INTERVENTION
- STORM RUNOFF → wadi discharge
- MOUNTAIN FRONT → alluvial fan

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FANS COALESCE → bajada; CLOSED BASIN → playa and salt concentration.

05

STAGE 05 THAR SPATIAL ANATOMY

THAR SPATIAL ANATOMY · SANDY MARUSTHALI CORE · BAGAR AND ARAVALLI TRANSITION · LUNI AND SALINE-BASIN ROUTE · CANAL-ALLUVIAL TRANSFORMATION

WEST → sandy Marusthali core

EAST → Bagar and Aravalli transition → SOUTHWEST → Luni and saline-basin route → NORTH → canal-alluvial transformation

SYSTEM / DEFINITION
- WEST → sandy Marusthali core
- EAST → Bagar and Aravalli transition

PROCESS / MECHANISM
- SOUTHWEST → Luni and saline-basin route
- NORTH → canal-alluvial transformation

SPATIAL / INDIA PATTERN
- WEST → sandy Marusthali core
- EAST → Bagar and Aravalli transition

HAZARD / LIMIT / RESPONSE
- SOUTHWEST → Luni and saline-basin route
- NORTH → canal-alluvial transformation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOUTHWEST → Luni and saline-basin route; NORTH → canal-alluvial transformation.

06

STAGE 06 SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND · NATURAL STRESS · COVER LOSS + SOIL DISTURBANCE · WATER MISMANAGEMENT · SALINITY OR DEPLETION · THRESHOLD CROSSED · RECOVERY CAPACITY DECLINES

VARIABLE RAIN + WIND → natural stress → COVER LOSS + SOIL DISTURBANCE → exposure → WATER MISMANAGEMENT → salinity or depletion → THRESHOLD CROSSED → recovery capacity declines

DEFINITION / COMPONENTS
- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

TRIGGER / MECHANISM
- WATER MISMANAGEMENT → salinity or depletion
- THRESHOLD CROSSED → recovery capacity declines

EFFECT / INTERVENTION
- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

07

STAGE 07 ARAVALLI BOUNDARY

ARAVALLI BOUNDARY · MODIFIES LOCAL WIND AND RUNOFF · DIVIDES AND ROUTES WATER · SUPPORTS CONNECTIVITY · NOT AN IMPERMEABLE DESERT WALL

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
RELIEF → modifies local wind and runoff	HABITAT → supports connectivity	RELIEF → modifies local wind and runoff
DRAINAGE → divides and routes water	LIMIT → not an impermeable desert wall	DRAINAGE → divides and routes water

DEFINITION / COMPONENTS
- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

TRIGGER / MECHANISM
- HABITAT → supports connectivity
- LIMIT → not an impermeable desert wall

EFFECT / INTERVENTION
- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HABITAT → supports connectivity; LIMIT → not an impermeable desert wall.

08

STAGE 08 CANAL BALANCE

CANAL BALANCE · WATER SUPPLY AND PRODUCTION · WATER-TABLE RISE · POOR DRAINAGE · WATERLOGGING AND SALINITY · CROP-WATER FIT + DRAINAGE + MONITORING

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
BENEFIT → water supply and production	POOR DRAINAGE → waterlogging and salinity	BENEFIT → water supply and production
SEEPAGE → water-table rise	MANAGEMENT → crop-water fit + drainage + monitoring	SEEPAGE → water-table rise

DEFINITION / COMPONENTS
- BENEFIT → water supply and production
- SEEPAGE → water-table rise

TRIGGER / MECHANISM
- POOR DRAINAGE → waterlogging and salinity
- MANAGEMENT → crop-water fit + drainage + monitoring

EFFECT / INTERVENTION
- BENEFIT → water supply and production
- SEEPAGE → water-table rise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGEMENT → crop-water fit + drainage + monitoring.

09

STAGE 09 LDN RESPONSE HIERARCHY

LDN RESPONSE HIERARCHY · PROTECT FUNCTIONING LAND · IMPROVE GRAZING, TILLAGE AND WATER USE · RESTORE SOIL, GRASS AND DRAINAGE · MONITOR GAINS AND LOSSES TRANSPARENTLY

AVOID → protect functioning land

REDUCE → improve grazing, tillage and water use → REVERSE → restore soil, grass and drainage → BALANCE → monitor gains and losses transparently

DEFINITION / COMPONENTS
- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

TRIGGER / MECHANISM
- REVERSE → restore soil, grass and drainage
- BALANCE → monitor gains and losses transparently

EFFECT / INTERVENTION
- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REDUCE → improve grazing, tillage and water use.

10

STAGE 10 RESTORATION MATRIX

RESTORATION MATRIX · WIND EROSION · COVER, SHELTER AND DUNE MANAGEMENT · RUNOFF EROSION · CONTOUR AND WATERSHED TREATMENT · DRAINAGE AND WATER BALANCE · RANGELAND DECLINE · NATIVE GRASS + INSTITUTIONS

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
WIND EROSION → cover, shelter and dune management	SALINITY → drainage and water balance	WIND EROSION → cover, shelter and dune management	SALINITY → drainage and water balance
RUNOFF EROSION → contour and watershed treatment	RANGELAND DECLINE → native grass + institutions	RUNOFF EROSION → contour and watershed treatment	RANGELAND DECLINE → native grass + institutions

CLASSIFICATION
- WIND EROSION → cover, shelter and dune management
- RUNOFF EROSION → contour and watershed treatment

DIAGNOSTIC BASIS
- SALINITY → drainage and water balance
- RANGELAND DECLINE → native grass + institutions

PROCESS LINK
- WIND EROSION → cover, shelter and dune management
- RUNOFF EROSION → contour and watershed treatment

EXAM TRAP
- SALINITY → drainage and water balance
- RANGELAND DECLINE → native grass + institutions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIND EROSION → cover, shelter and dune management.

11

STAGE 11 DESERTIFICATION ANSWER SPINE

DESERTIFICATION ANSWER SPINE · RETAIN UNCCD DRYLAND BOUNDARY · LOCATE PROCESS AND PRESSURE · SUSCEPTIBILITY PLUS ACCELERATOR · PROCESS-MATCHED, MONITORED AND QUALIFIED

DEFINE → retain UNCCD dryland boundary → MAP → locate process and pressure → DIAGNOSE → susceptibility plus accelerator → RESPOND → process-matched, monitored and qualified

DEFINITION / COMPONENTS
- DEFINE → retain UNCCD dryland boundary
- MAP → locate process and pressure

TRIGGER / MECHANISM
- DIAGNOSE → susceptibility plus accelerator
- RESPOND → process-matched, monitored and qualified

EFFECT / INTERVENTION
- DEFINE → retain UNCCD dryland boundary
- MAP → locate process and pressure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIAGNOSE → susceptibility plus accelerator; RESPOND → process-matched, monitored and qualified.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE THAR LIES WEST OF THE ARAVALLI, BETWEEN THE · BOOK ESTIMATES OF DESERT AREA VARY WITH THE BOUNDARY · THE RANN OF KACHCHH IS TREATED AS AN EXTENSION · ECONOMY (MAJID) · BAJRA, PULSES, FODDER; LIVESTOCK IS CENTRAL; SCOPE FOR DATE-PALM · IT IS DESCRIBED AS THE WESTERN DRY REGION — · ~29.7% OF INDIA (~97.85 MHA) IS DEGRADED (2018-19), UP

OPTIONAL PROCESS DEPTH
- The Thar lies west of the Aravalli, between the Indus and the Aravalli, spilling into Punjab, Haryana and northern Gujarat.
- Book estimates of desert area vary with the boundary used; retain them as source-era figures, not
- The Rann of Kachchh is treated as an extension of the desert; Jaisalmer and Barmer show the strongest aeolian (dune) topography.
- Economy (Majid): bajra, pulses, fodder; livestock is central; scope for date-palm, water-melon and rain-water harvesting.

DATA / INSTITUTION LIMIT
- It is described as the Western Dry Region — an arid tract (Marwar/Mewar margins) with extreme temperature range.
- ~29.7% of India (~97.85 mha) is degraded (2018-19), up from 28.76% (2003-05) — a rising trend.
- Rajasthan has the largest desertified area; wind erosion drives 40%+ of its desertification.
- India targets Land Degradation Neutrality (LDN) by 2030 under the UNCCD; tools include Green India Mission and watershed/afforestation programmes.

CONTEMPORARY PARALLEL
- Thar = west of Aravalli, between Indus & Aravalli.
- Area figures vary by delimitation; western core is arid and margins semi-arid.
- Rann of Kachchh is linked to the wider arid region but is a distinct saline mudflat/marsh.
- Jaisalmer & Barmer = strongest aeolian (dune) topography.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels